

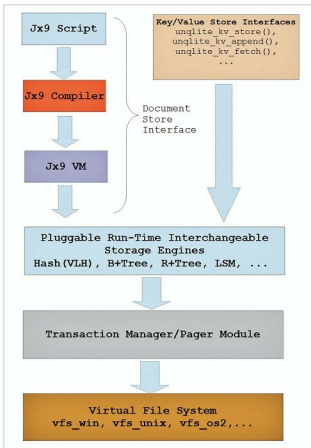


Figure 2: UnQLite architecture

Berkeley DB

Berkeley DB is an open source, high performance, industry leading embeddable storage engine that provides developers a fast, reliable, local database with almost zero administration. It was primarily developed by Sleepycat Software but was acquired by Oracle in 2006. Berkeley DB is written in C language with API bindings for C++, C#, Java, Perl, PHP, Python, Ruby, Smalltalk, TCL and many other programming languages. It provides powerful APIs for many languages including Android and iOS.

Berkeley DB provides powerful robust data storage features like other traditional RDBMSs such as ACID transactions, and recovery, locking, multi-process and multi-threading for high concurrency, replication and hot and cold backup.

Oracle Berkeley DB: Berkeley DB has powerful architecture, which is very simple compared to other RDBMS software. Oracle added support for SQL in the 11g R2 release based on the popular SQLite API by including a version of SQLite in Berkeley DB. All the powerful features of SQLite are included with Berkeley DB. As it includes the fully-featured SQL database engine, you can use JDBC, ODBC or any other compatible access layer as well.

Berkeley DB is highly flexible, and you can use it across a plethora of applications and as a replacement for custom, home-grown solutions. It provides fast, reliable and scalable persistence for applications that need to store data locally, run without interruption and access data in a predictable fashion.

Editions

The following are the three editions of Oracle Berkeley DB:

- Oracle Berkeley DB: Written in C. Current version: 6.2
- Berkeley DB Java Edition: Written in Java. Current version: 7.5.11
- Berkeley DB XML: Written in C++. Current version: 6.1.4

Features

- **Data management:** Fine grained and configurable locking for concurrent systems; sequential and indexed retrieval; stores data in-memory or on disk or a combination of both; strong encryption; library for data management to link embedded processes; and highly portable.
- **Transactions:** ACID compliant, nested transactions, disaster recovery options, configurable graph, deadlock detection and log file archival.
- **Replication:** Single and multiple replica models; delayed client synchronisation; scalable to thousands of replica nodes; non-stop upgrades; and PAXOS-compliant election algorithm.
- **Deployment:** Cross-platform, binary installer, open source and has a powerful administration.

Official website: <http://www.oracle.com/technetwork/database/database-technologies/berkeleydb/overview/index.html>

Latest version: Berkeley DB 12Cr1 (12.1.6.2.32)

ObjectBox DB

ObjectBox DB, developed by Greenrobot, is an object-oriented embedded database and is regarded as the best alternative to SQLite for mobile databases. It is specially designed for mobile database management and brings NoSQL Server technology to mobiles.

The guiding principle behind the ObjectBox DB design is, 'Less code is better code', which means ObjectBox DB is easy to work with even for newbies.

ObjectBox is a fully transactional database and consists of all the ACID features. It is a super-fast mobile database that persists objects and enables users to avoid repetitive tasks, providing a simple interface to data.

The ObjectBox database extensions combine advanced enterprise application capabilities — self-adjusting connection pooling, transaction handling and RDB-to-XML mappings — with a really straightforward o:XML interface.

ObjectBox DB is a fast, object-oriented mobile database for Android and Linux, and soon will support iOS. It supports JavaRx and Kotlin.